

MODELING CHLORINE RESIDUALS IN DRINKING-WATER DISTRIBUTION SYSTEMS

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ABSTRACT: A mass-transfer-based model is developed for predicting chlorine decay in drinking-water distribution networks. The model considers first-order reactions of chlorine to occur both in the bulk flow and at the pipe wall. The overall rate of the wall reaction is a function of the rate of mass transfer of chlorine to the wall and is therefore dependent on pipe geometry and flow regime. The model can thus explain field observations that show higher chlorine decay rates associated with smaller pipe sizes and higher flow velocities. It has been incorporated into a computer program called EPANET that can perform dynamic water-quality simulations on complex pipe networks. The model is applied to chlorine measurements taken at nine locations over 53 h from a portion of the South Central Connecticut Regional Water Authority's service area. Good agreement with observed chlorine levels is obtained at locations where the hydraulics are well characterized. The model should prove to be a valuable tool for managing chlorine-disinfection practices in drinking-water distribution systems.

INTRODUCTION

Chlorine is widely used as a disinfectant in drinking-water systems throughout the world. Most water suppliers attempt to maintain a detectable chlorine residual within the distribution system to minimize the potential for microbial growth. As chlorine travels through the pipes in a distribution system it can react with a variety of materials both within the bulk water and from the pipe wall. Additional reactions can occur in storage facilities. For any parcel of water, these reactions decrease its chlorine content, depending on its travel time through the pipe network and its residence time in storage tanks. An ability to understand these reactions and model their impact throughout a distribution system will assist water suppliers in selecting operational strategies and capital improvements to insure delivery of high-quality drinking water.

Clark et al. (1993) showed how chlorine residuals can vary throughout the day at different locations in a distribution system depending on the flow path and residence time of the water reaching a location. Studies on chlorine decay rates in single lengths of pipe reveal that the decay rate in the pipe is several times greater than the decay rate of the same water in a flask (Wable et al. 1991). This suggests that the pipe wall is somehow contributing to the overall chlorine demand observed in distribution systems. Hunt and Kroon (1991) described a network model for chlorine residual that used a first-order decay reaction with a unique rate constant for each pipe and

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storage facility. In calibrating their model, they noted that smaller pipes off of the main transmission lines required larger decay-rate constants to match observed chlorine levels better. This is consistent with the fact that smaller pipes offer a larger wall-surface area per unit of flow volume for reaction to occur. Biswas et al. (1993) developed a model for chlorine decay within single pipes under steady-state flow conditions that included both bulk-flow reaction and radial diffusion, and subsequent pipe-wall reaction of chlorine.

The present paper develops a mass-transfer-based model of chlorine decay in pipe networks that applies to nonsteady flow under both turbulent and laminar conditions. The model has been incorporated into a general-purpose distribution-system water-quality simulation program called EPANET. The present paper discusses how EPANET was used to calibrate the model to field observations taken from a portion of the South Central Connecticut Regional Water Authority (SCCRWA) and reports on its ability to match measured changes in chlorine levels throughout the system over time.

MODEL DEVELOPMENT

Based on previous work (Wable et al. 1991; Sharp et al. 1991), it appears reasonable to assume that the disappearance of chlorine flowing through a pipe is governed by first-order kinetics. We assume that this disappearance is due to reactions both within the bulk flow and at sites along the pipe wall (or in close proximity to the wall). The rates of these reactions can be different, with the overall rate of the wall reaction also being affected by the rate at which chlorine can be transported from the bulk flow to the pipe wall. We assume that this latter step can be adequately represented by a film resistance model of mass transfer using a mass-transfer coefficient suitable to the flow regime in the pipe. An alternative mechanism of wall-related reaction, namely the transport of reactive species off of the wall into the bulk flow, is not considered here as it demands significantly more information from which to build a model.

With these assumptions, the one-dimensional conservation-of-mass equation for a dilute concentration of total free chlorine in water flowing through a section of a pipe is

$$\frac{\partial c}{\partial t} = -u \frac{\partial c}{\partial x} - k_b c - \frac{k_f}{r_h} (c - c_w) \quad (1)$$

where c = chlorine concentration in the bulk flow; t = time; u = flow velocity in pipe; x = distance along pipe; k_b = decay rate constant in the bulk flow; k_f = mass-transfer coefficient; r_h = hydraulic radius of pipe (one half the pipe radius); and c_w = chlorine concentration at the pipe wall.

The term on the left side of (1) represents the rate of change of chlorine concentration within a differential section of pipe. The first term on the right side of the equation accounts for the advective flux of chlorine through the section (dispersive flux is assumed to be negligible under typical operating conditions). The second term represents chlorine decay within the bulk flow, and the third term accounts for transport of chlorine from the bulk flow to the pipe wall and subsequent reaction. The inverse of the hydraulic radius represents the specific surface area (i.e., the pipe-wall area per unit of pipe volume) available for reaction.

Assuming that the reaction of chlorine at the pipe wall is first-order with respect to the wall concentration c_w and that it proceeds at the same rate

as chlorine is transported to the wall (so there is no buildup of chlorine at the wall) results in the following mass balance for chlorine at the wall:

$$k_f(c - c_w) = k_w c_w \quad (2)$$

where k_w = a wall decay constant with units of length over time. Solving (2) for c_w and substituting it into (1) gives

$$\frac{\partial c}{\partial t} = -u \frac{\partial c}{\partial x} - k_b c - \frac{k_w k_f c}{r_h(k_w + k_f)} \quad (3)$$

Standard literature expressions can be used for the mass-transfer coefficient k_f (Edwards et al. 1976):

$$k_f = \text{Sh} \frac{D}{d} \quad (4)$$

$$\text{Sh} = 0.023 \text{Re}^{0.83} \text{Sc}^{0.333}, \quad \text{for } \text{Re} > 2,300 \quad (5)$$

$$\text{Sh} = 3.65 + \frac{0.0668(d/L)(\text{Re Sc})}{1 + 0.04[(d/L)(\text{Re Sc})]^{2/3}}; \quad \text{for } \text{Re} < 2,300 \quad (6)$$

$$\text{Re} = \frac{ud}{\nu} \quad (7)$$

$$\text{Sc} = \frac{\nu}{D} \quad (8)$$

where Sh = Sherwood number; Re = Reynolds number; Sc = Schmidt Number; D = molecular diffusivity of chlorine in water; ν = kinematic viscosity of water; d = pipe diameter; and L = pipe length. Note that for a particular chemical species, k_f is a function of pipe diameter, flow velocity, and temperature (as it affects diffusivity and viscosity).

Eq. (3) describes the time variation of chlorine along a single pipe. For a distribution system, such as shown in Fig. 1, the mass-conservation equation for the i th pipe can be expressed

$$\frac{\partial c_i}{\partial t} = -u_i \frac{\partial c_i}{\partial x_i} - K_i c_i \quad (9)$$

where subscript i indicates the i th pipe in the network; and K = an overall decay constant that contains the bulk decay constant, the hydraulic radius, the mass-transfer coefficient, and the wall decay constant as follows:

$$K_i = k_b + \frac{k_w k_f}{r_{hi}(k_w + k_f)} \quad (10)$$

For known system hydraulics (which may change over time), (9) can be solved with a known initial condition for chlorine throughout the network at time 0 and a boundary condition at the head end junction of each pipe i where $x_i = 0$. Assuming that complete and instantaneous mixing occurs at pipe junctions, this boundary condition can be expressed with the following conservation-of-mass equation:

$$c_i|_{x=0} = \frac{\sum q_k c_k|_{x=L} + M_i}{\sum q_k + S_i} \quad (11)$$

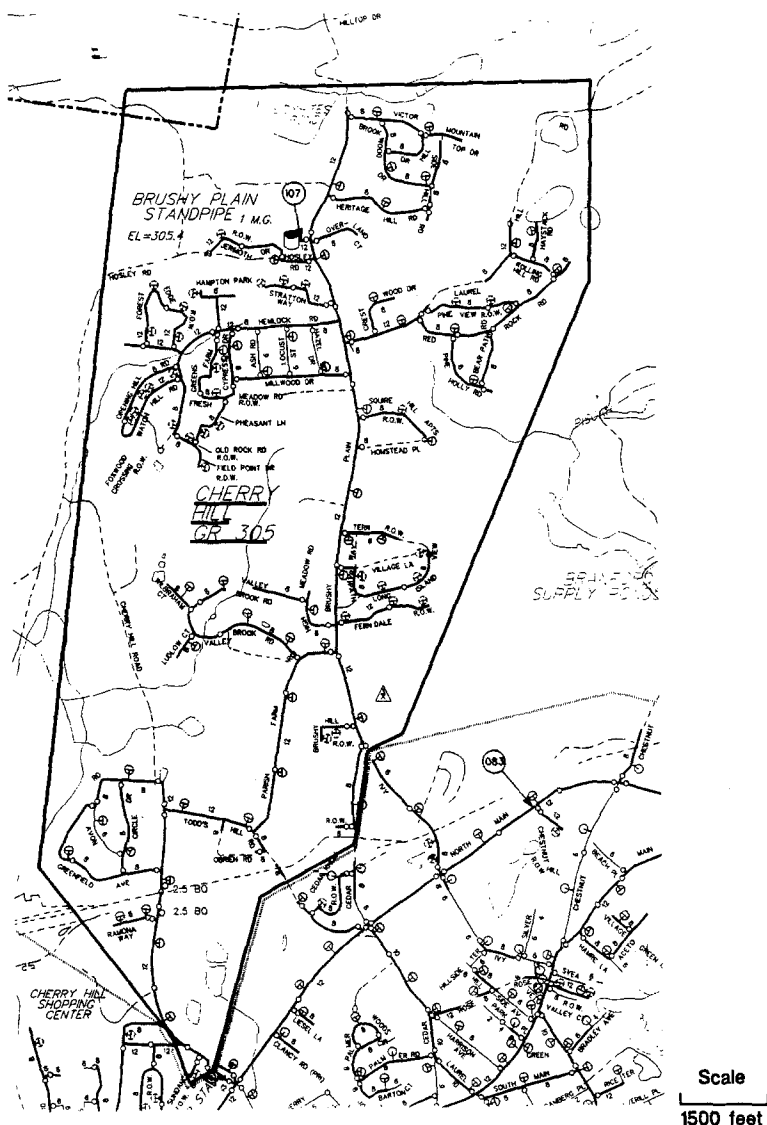


FIG. 1. The Cherry Hill/Brushy Plains Study Area

The summation is made over all pipes k that have flow q_k into the head junction of pipe i ; M_i = any external mass flow of chlorine introduced at the head of pipe i ; and S_i = any external flow of water introduced at the head of pipe i . Note that to solve for the chlorine concentration within pipe i , one first has to know what the concentrations are in all pipes flowing into pipe i .

Storage tanks can be modeled as completely mixed, variable-volume reactors where the changes in volume and concentration over time are

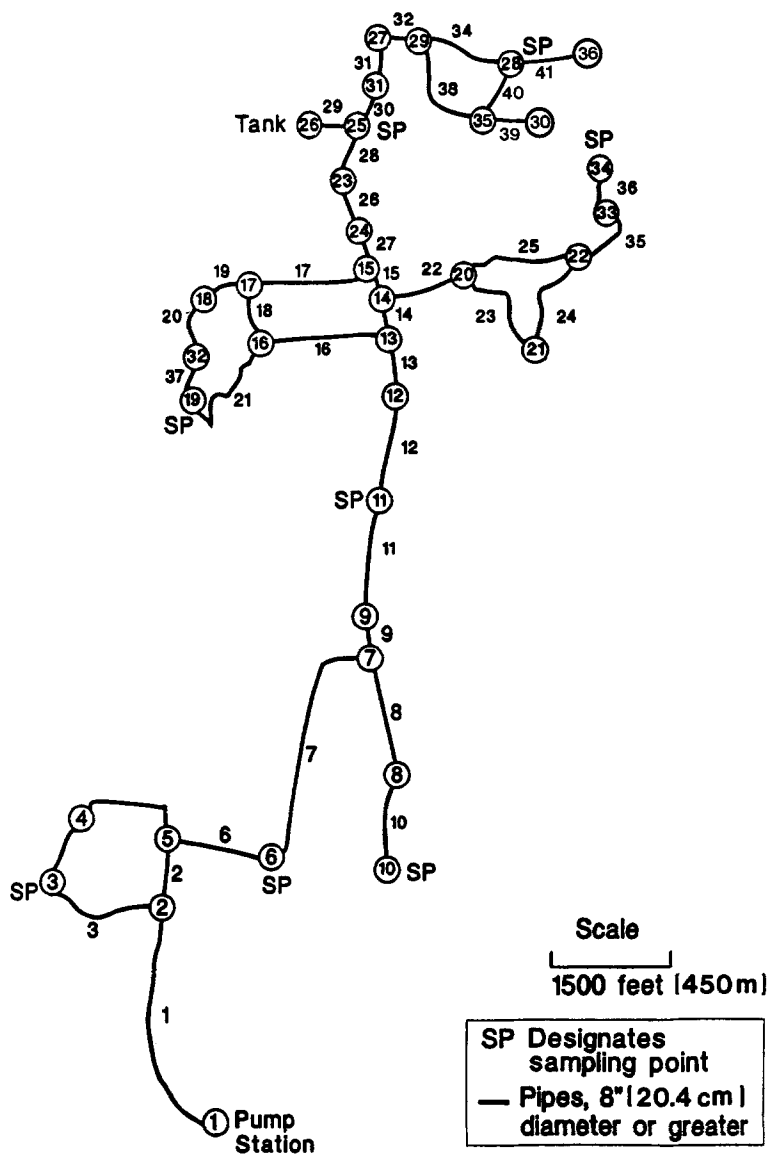


FIG. 2. Node-Link Representation of Study Area

$$\frac{\partial V}{\partial t} = \sum_{k \in I} q_k - \sum_{j \in O} q_j \quad (12)$$

$$\frac{\partial(Vc)}{\partial t} = \sum_{k \in I} q_k c_k - \sum_{j \in O} q_j c - k_b c \quad (13)$$

where V = tank volume; I = set of pipes with flow into the tank; and O = set of pipes with flow out of the tank.

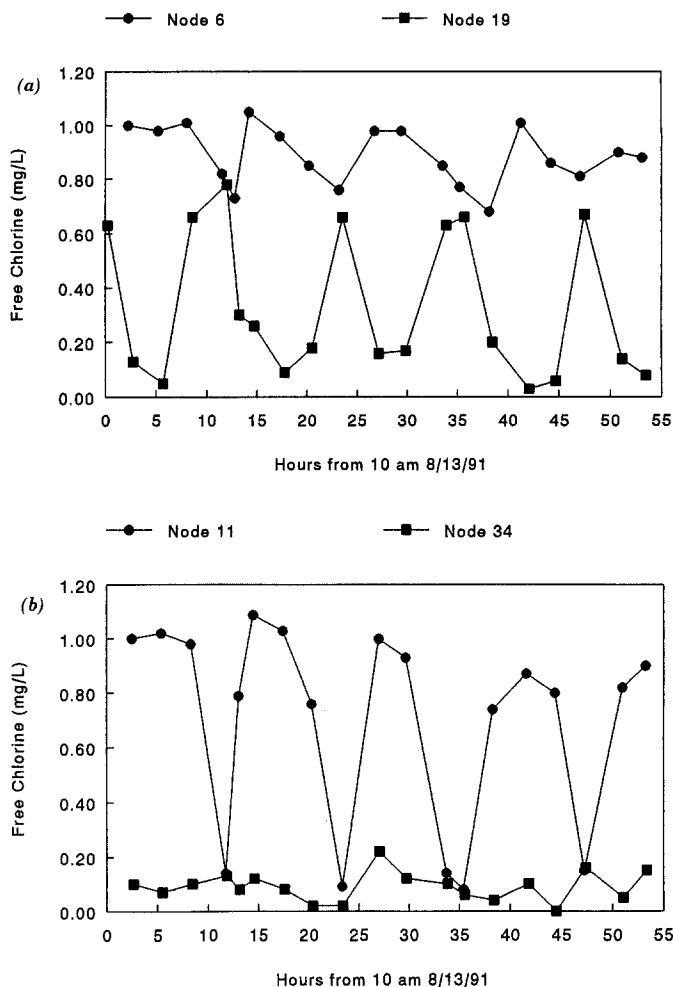


FIG. 3. Residual Free Chlorine Concentrations Observed on August 13–15, 1991, at: (a) Nodes 6 and 19; and (b) Nodes 11 and 34

Eqs. (9)–(13) represent a coupled set of differential/algebraic equations over all pipes in the network. Under a set of known, time-varying hydraulic conditions, they can be solved using an explicit discretization technique called the Discrete Volume Element Method (DVEM) (Rossman et al. 1993). Within each time period when hydraulic conditions are constant, DVEM divides each pipe into a number of segments derived from the pipe's volume, its flow velocity, and a time step equal to the shortest time of travel through any pipe. Each segment is treated as a completely mixed reactor. At each time step, the mass contained in each pipe segment is first reacted and then transferred to the adjacent downstream segment. When the adjacent segment is a junction node, the mass and flow entering the node is added to any mass and flow already received from other pipes. After these reaction/transport steps are completed for all pipes, the resulting mixture concentration at each junction node is computed and then released into the

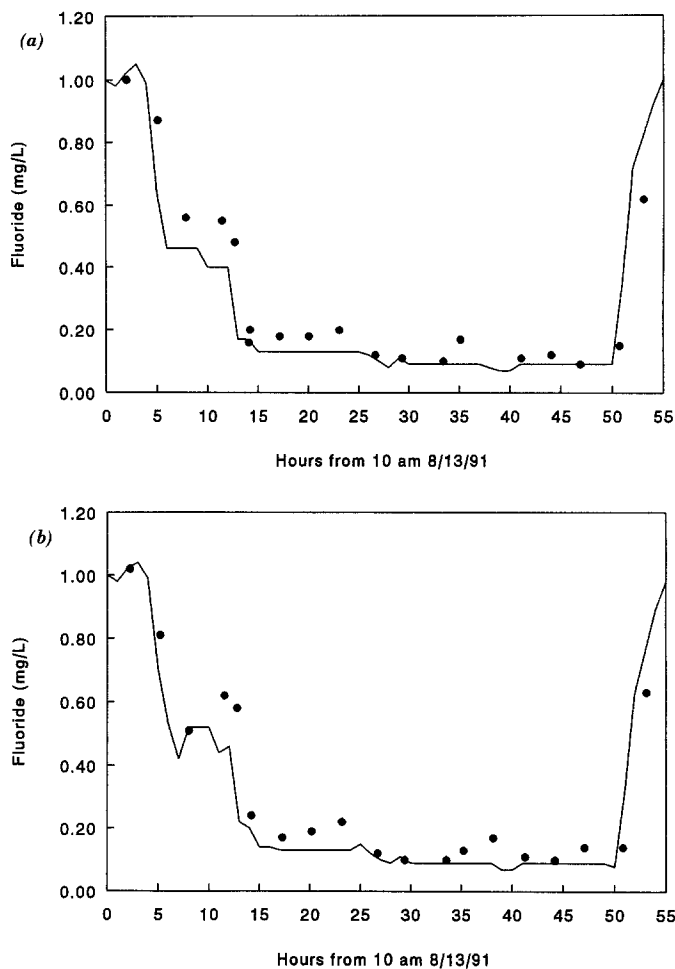


FIG. 4. Comparison between Predicted (Solid Line) and Observed (Circle) Fluoride Concentrations at: (a) Location 3; and (b) Location 6

head end segments of pipes with flow leaving the node. This sequence of steps is repeated until the time when a new hydraulic condition occurs. The network is then resegmented, and the computations are continued.

DVEM has been incorporated into a general-purpose distribution-system simulation computer code called EPANET (Rossman 1994). EPANET performs both extended-period-hydraulic and water-quality simulations. In addition to chemical propagation, it can also perform dynamic simulations of water age and track the percentage of flow received from any particular source.

MODEL APPLICATION

On August 13–15, 1991, a sampling study in the Cherry Hill/Brushy Plains Service Area of the SCCRWA was undertaken. The results from this study

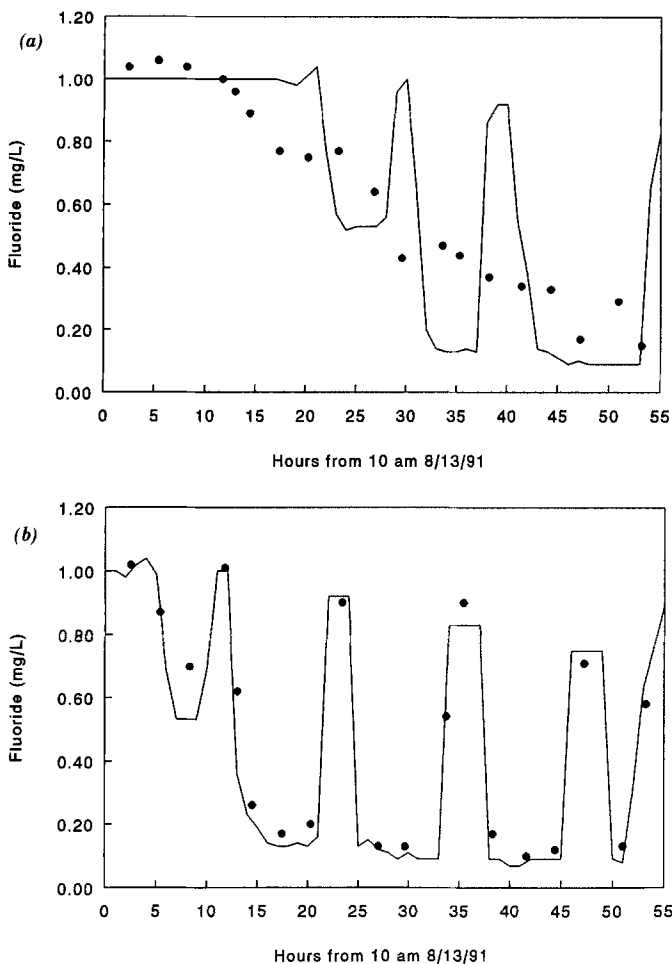


FIG. 5. Comparison between Predicted (Solid Line) and Observed (Circle) Fluoride Concentrations at: (a) Location 10; and (b) Location 11

have been used to characterize chlorine decay within single lengths of pipe (Biswas et al. 1993), in pipe networks (Clark et al. 1993) and in storage tanks (Grayman and Clark 1993). We build on this prior work by applying our mass-transfer-based chlorine decay model to these same data. To provide a sufficiently clear description of the study and its results, we must reiterate some of the material described in these latter references.

The service area sampled covers approximately 5.2 km² (2 sq mi) and is almost entirely residential. Average water use during the sampling period was 20.2 L/s (0.46 mgd). The distribution system is composed of 20.3 cm (8 in.) and 30.5 cm (12 in.) mains as shown in Fig. 1. It receives its water from SCCRWA's Saltonstall treatment plant. Water is pumped into the service area at the Cherry Hill Pump Station, and storage is provided at the other end of the service area by the Brushy Plains Tank. The pump station has a capacity of 61.3 L/s (1.4 mgd), and the tank's capacity is 3,800,000 L

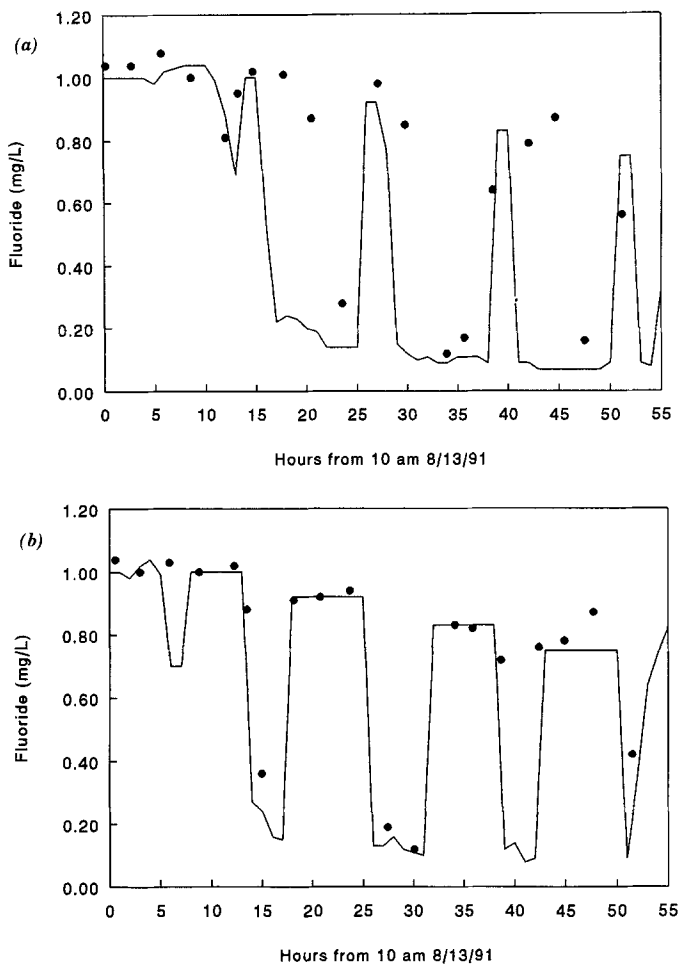


FIG. 6. Comparison between Predicted (Solid Line) and Observed (Circle) Fluoride Concentrations at: (a) Location 19; and (b) Location 25

(1,000,000 gal.). During normal operation, pumping occurs when the water level in the tank drops to 17.1 m (56 ft) and ceases when the water level reaches 19.8 m (65 ft).

Sampling was conducted at the pump station and eight locations throughout the service area as shown in Fig. 2. The pump station was sampled on its discharge side. The tank was sampled on a common inlet/outlet line. During the tank fill cycle, a sample from this line represented water entering the tank, but during the emptying cycle it represented the tank contents. The remaining seven locations were at hydrants.

At 9:00 a.m. on August 13, the fluoride feed was shut off at the Saltonstall treatment plant. This allowed fluoride to be used as a conservative tracer for network model calibration. Beginning at this time, a "circuit" was begun wherein samples were taken at each of the nine locations in the service area. The sampling circuit took approximately 1.5–2 h to complete, and a

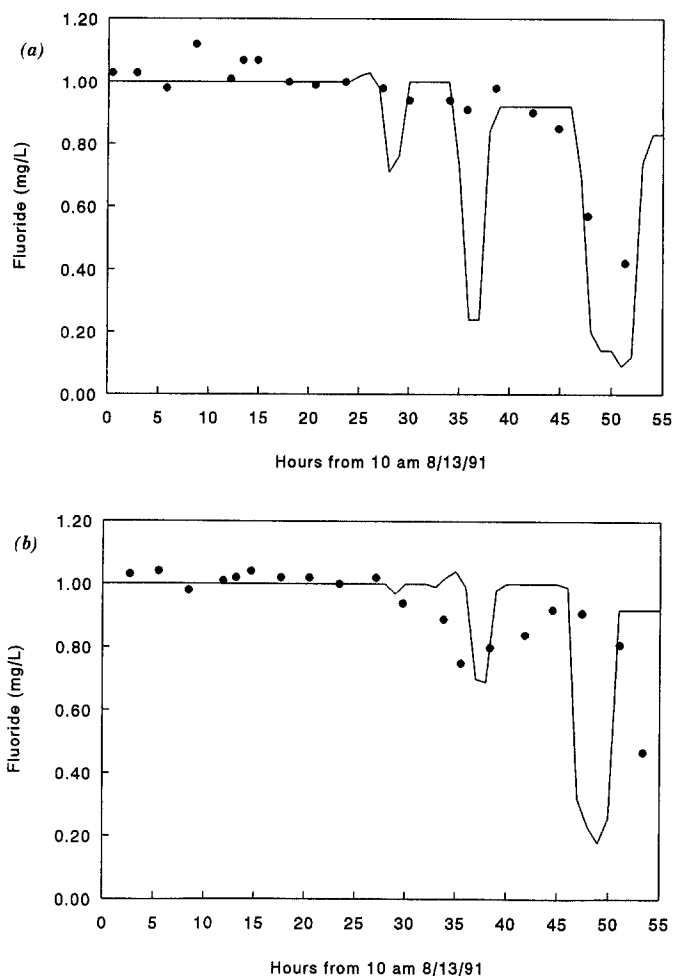


FIG. 7. Comparison between Predicted (Solid Line) and Observed (Circle) Fluoride Concentrations at: (a) Location 28; and (b) Location 34

new circuit was begun every 3 h or so. The total duration of the sampling was 53 h.

Free available chlorine at the pump station and at the tank was determined using continuous chlorine analyzers (Rosemount Model 4024, Santa Clara, Calif.) equipped with a free chlorine-specific membrane electrode (Rosemount Model 90243-116). Grab samples from other locations were analyzed for free available chlorine in the field with a portable DPD colorimetric test kit (Hach Model 46700-05, Loveland, Colo.). Continuous fluoride monitoring was conducted at the tank using an ion-selective electrode analyzer (Orion Model 720, Boston, Mass.), whereas grab samples collected at other locations were analyzed for fluoride at the SCCRWA laboratory using a standard ion-selective electrode method. Over the 53 h sampling period, 181 pairs of chlorine and fluoride values were obtained for the nine sites sampled.

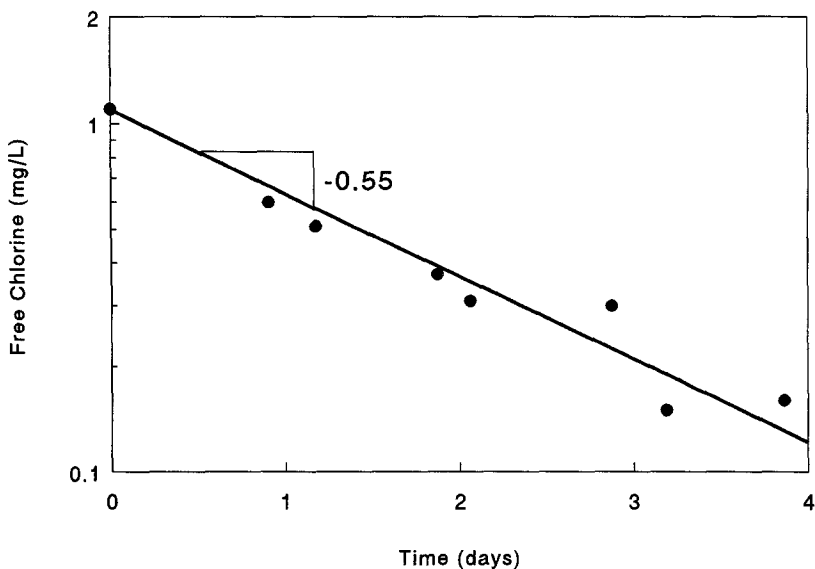


FIG. 8. Results of Free Chlorine Decay Bottle Test

ANALYSIS OF RESULTS

A general picture of chlorine residual behavior in the Cherry Hill/Brushy Plains system can be seen in Fig. 3. Time-series plots of observed chlorine levels are shown for several sampling locations. The cyclical behavior of the chlorine residuals appears to follow the pumping pattern. High residuals occur at times when "new" water with high chlorine levels is pumped into the system; low residuals occur when the pumps are off and the system is being fed by "old" water from the tank. This behavior is not as apparent at the spurs of the system where a distinction between "new" and "old" water may not be as clear.

The mass-transfer-based chlorine decay model, as incorporated in the EPANET program, was applied to the data collected from the Cherry Hill/Brushy Plains service area. For modeling purposes, the service area was represented with the pipe network shown in Fig. 2. It included all 30.5 cm (12 in.) mains, major 20.3 cm (8 in.) mains and loops, and all pipes connected to the sampling sites. Pipe lengths were scaled from maps, actual pipe diameters were used, and all pipes were initially assigned a Hazen-Williams roughness coefficient of 100. Initial estimates of average water-use rates throughout the service area were based on the type and number of housing units surrounding each junction node in the network. Total water usage in the service area over any interval of time could be computed from SCCRWA's continuous recording of flow at the Cherry Hill pump station and water levels in the Brushy Plains tank.

The hydraulic model of the service area and its calibration were discussed in Clark et al. (1993). Extended-period hydraulic simulation was used to compute flows and velocities in all pipes in the network at 1 h intervals over the 53 h sampling period. The principal adjustable parameters in the simulation, the average water-use rates at the pipe junctions, and several roughness coefficients were adjusted so that the simulated transport of fluoride throughout the network over the 53 h sampling period gave a good match

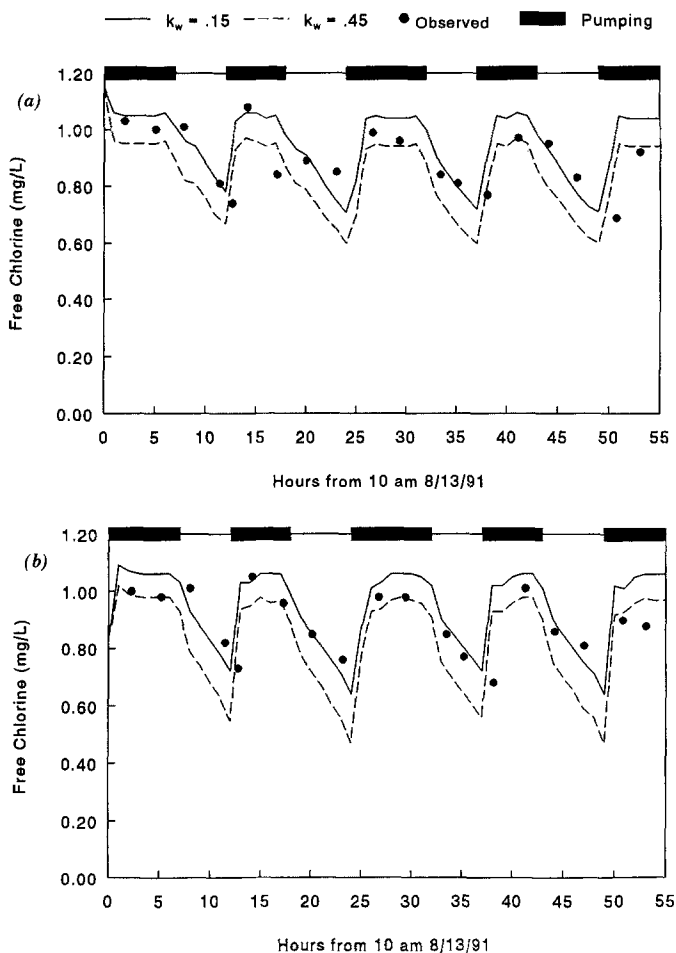


FIG. 9. Comparison between Predicted and Observed Chlorine Concentrations at: (a) Location 3; and (b) Location 6

to the observed fluoride readings. Figs. 4–7 show the results achieved from the hydraulic calibration.

The parameters of the calibrated hydraulic model were then used to prepare an input-data set for the EPANET program so that free chlorine residuals could be modeled. This data set also included estimates of initial free chlorine residuals at the junction nodes of the network. These were developed from initial measurements taken during the field sampling. The free chlorine content of water introduced into the service area at the pump station was kept constant at 1.1 mg/L. The only remaining parameters to be specified in the chlorine decay model were the bulk and wall decay constants (k_b and k_w , respectively). Although it is conceivable that k_b and k_w could vary from pipe to pipe, in the spirit of model parsimony they were kept constant. In this case, any variation in reaction rates at different points of the network would be due only to differences in chlorine concentration, pipe size, and flow velocity.

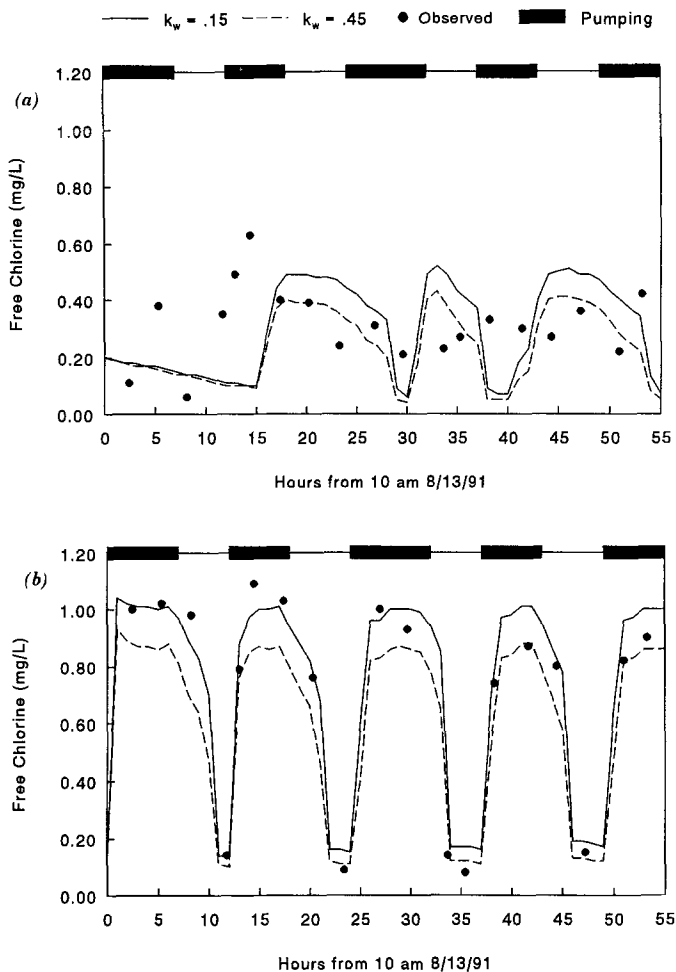


FIG. 10. Comparison between Predicted and Observed Chlorine Concentrations at: (a) Location 10; and (b) Location 11

The bulk decay constant k_b in all pipes and in the tank was assigned a value of 0.55 days^{-1} based on a laboratory beaker test of free chlorine decay in water taken from the service area. The test was performed by SCCRWA's laboratory. Multiple 575 ml bottles were filled with effluent from the Saltonstall treatment plant and stored at constant temperature. Twice a day a bottle was tested for free available chlorine and then discarded. Fig. 8 plots the observed decay in chlorine residual over time. The best-fit first-order decay constant for these data was 0.55 days^{-1} .

The remaining parameter k_w was adjusted over a range of values, and the simulated EPANET results were compared to the observed data at the eight sampling locations out in the network. Figs. 9–12 show the results obtained for a range of k_w values between 0.15 m/day (0.5 ft/day) and 0.45 m/day (1.5 ft/day). Most of the observations at nodes 3, 6, 11, 19, and 25 fall within the simulated range between these two k_w values. The higher

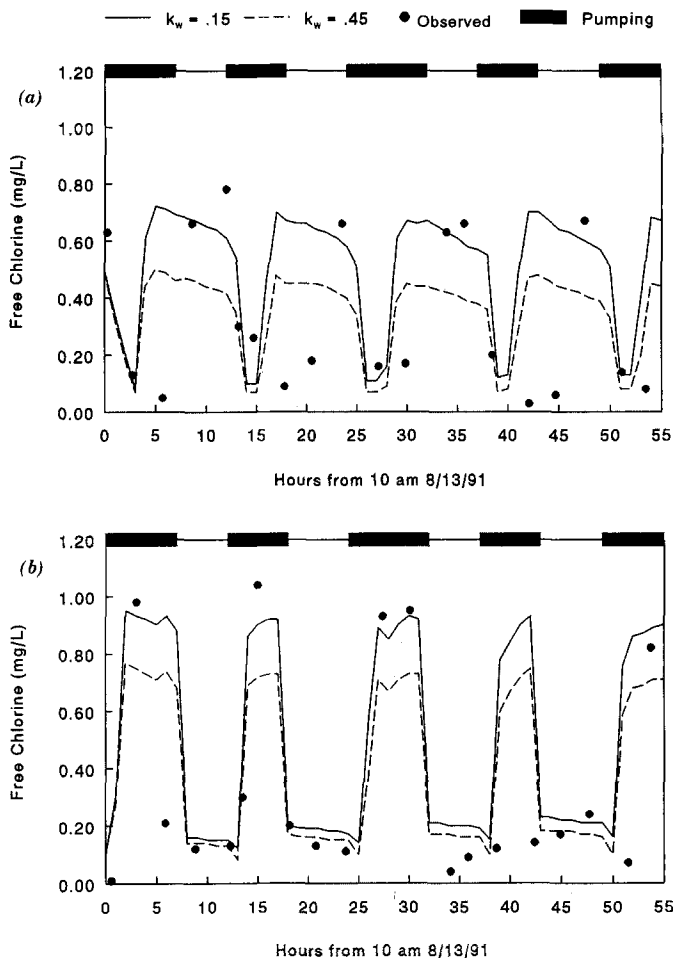


FIG. 11. Comparison between Predicted and Observed Chlorine Concentrations at: (a) Location 19; and (b) Location 25

k_w value has a smaller root-mean-square residual error (0.186 mg/L) than the lower k_w (0.211 mg/L), but does not do as well in matching the chlorine peaks at nodes 11, 19, and 25.

The wide fluctuations in chlorine at locations such as node 25 reflect the reversal in flow as the system is fed either with “new” water from the pump or “old” water from the tank. Note that the chlorine decay experienced by this “old” water is more a function of the average residence time in the tank (about four days) than the average cycle time for the pump (about 12 h). The model overpredicts chlorine residuals at nodes 10, 28, and 34. These nodes are at sparsely populated dead ends where accurate estimation of water usage over time was particularly difficult. Indeed, Figs. 4–7 reveal that these same nodes had the poorest hydraulic calibration.

It is informative to compare the relative contributions of the bulk decay constant k_b and the wall decay constant k_w to the overall decay constant K . In a pipe with r_h of 1 m and high flow velocity (so mass transfer is not

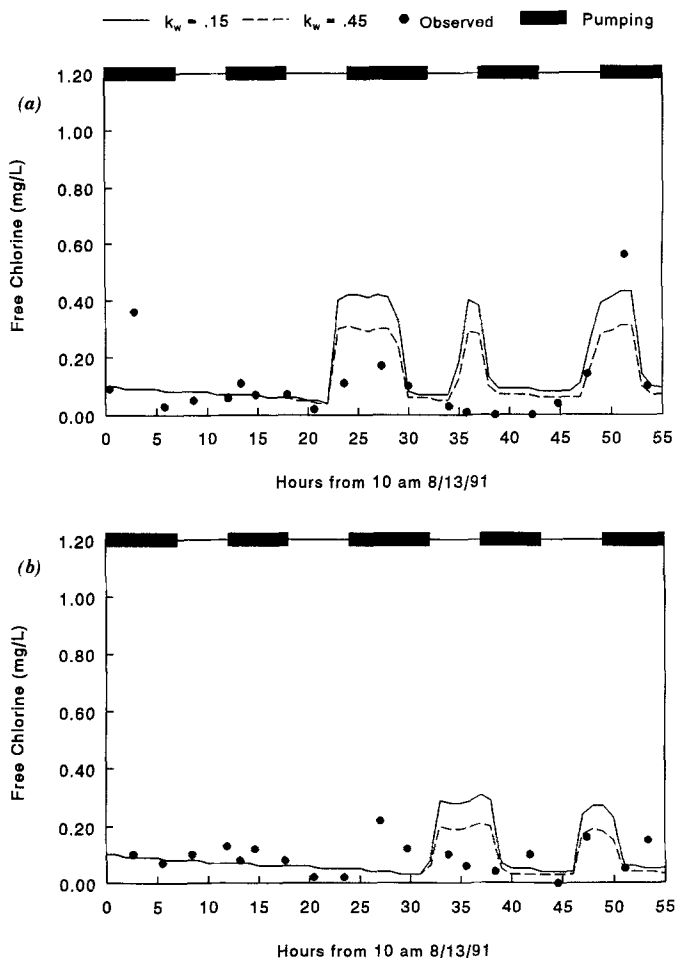


FIG. 12. Comparison between Predicted and Observed Chlorine Concentrations at: (a) Location 28; and (b) Location 34

a rate limiting factor), k_b and k_w contribute equally to K [see (10)]. As velocity becomes smaller, the contribution from k_w is decreased. As the pipe diameter is made smaller, the influence from k_w is increased. Fig. 13 illustrates this behavior for the pipe sizes and flow velocities as well as for the bulk and wall decay constants used for the Cherry Hill/Brushy Plains network. At the lower wall decay constant of 0.15 m/day, the curves for both the 20.3 cm (8 in.) and 30.5 cm (12 in.) pipes remain relatively flat over a wide range of velocities. This indicates that the overall decay rate of chlorine is wall rate limited, not mass-transfer limited. At the higher wall decay constant of 0.45 m/day, the opposite appears to be the case.

Fig. 14 shows the total loss rate of chlorine in the system due to reactions in the tank, in the bulk flow, and at the pipe walls. The higher wall decay constant of 0.45 m/day was used to develop this plot. The loss along the pipe wall clearly dominates the other two losses. On average, the wall accounts for 67% of chlorine losses, the bulk flow for 12%, and the tank

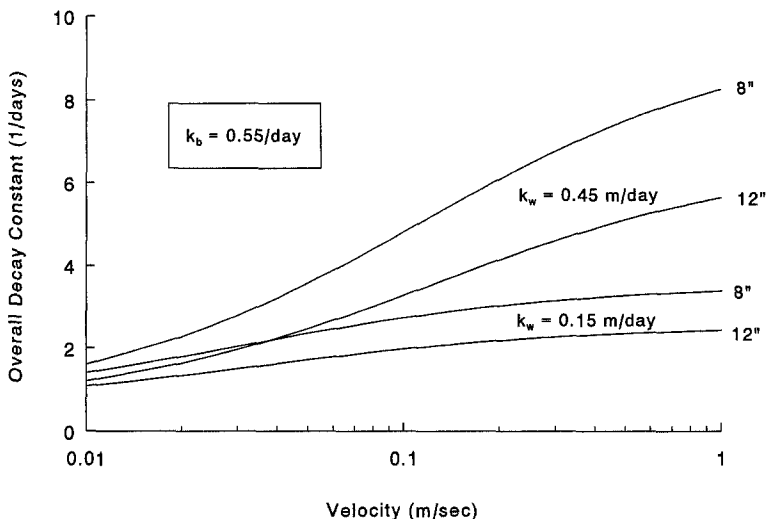


FIG. 13. Dependence of Overall Decay Constant on Pipe Diameter and Flow Velocity

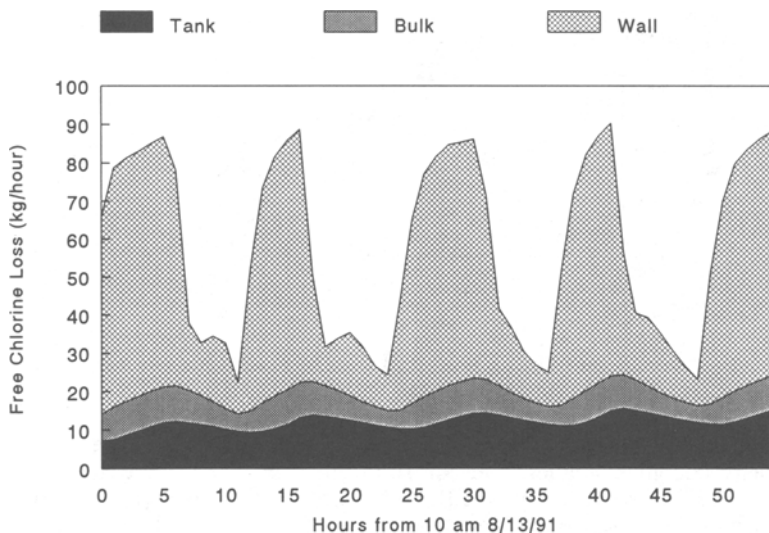


FIG. 14. Predicted Free Chlorine Loss Rates in Study Area

for 21%. Similar behavior is observed at the lower wall decay constant (48% loss at the wall, 19% in the bulk flow, and 33% in the tank). The periods of increased wall demand occur when the pump is on and velocities are high, thus leading to high wall reaction rates. When the pump is off, the reverse occurs. One conclusion to be drawn from this plot is that changes in tank operation to reduce the holding time of its contents would have limited effect in reducing overall chlorine losses in this system. A pipe cleaning or replacement program might be more effective.

SUMMARY AND CONCLUSIONS

A mass-transfer-based model of chlorine decay in water-distribution networks has been developed. The model is capable of explaining observed phenomena in past chlorine decay studies, such as higher decay rates in smaller pipes or in higher velocity flows. Aside from initial conditions, it contains two sets of rate coefficients that must be estimated. One represents a rate constant for reactions occurring in the bulk flow, and the other is a pipe-wall reaction-rate constant.

The chlorine decay model was applied to data collected over a 53 h sampling period in a portion of the SCCRWA. The model's bulk decay-rate constant was determined independently in the laboratory. Its wall decay constant was varied over a range of values that included both reaction-rate-limiting and mass-transfer-rate-limiting values. The resulting predictions were compared with observed chlorine measurements at eight locations. Good agreement was achieved at locations where the hydraulic conditions were well characterized. Model predictions were less accurate at sites where the hydraulic calibration was less successful. These results underscore the need to obtain accurate hydraulic information before running a network water-quality model.

Use of a single wall decay constant for all pipes in a network might not be appropriate for all distribution systems. A fertile area for future research is to understand how this constant might depend on pipe characteristics, such as age or material, and on water characteristics indicative of biofilm occurrence or corrosion activity. The chlorine decay model and its implementation within the EPANET program will provide water-supply managers with a useful tool for understanding the behavior of chlorine residuals throughout their systems under a wide variety of changing hydraulic conditions.

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APPENDIX II. NOTATION

The following symbols are used in this paper:

- c = bulk flow concentration (ML^{-3});
- c_w = wall concentration (ML^{-3});
- D = molecular diffusivity (L^2T^{-1});
- d = pipe diameter (L);
- I = set of pipes with flow into junction;
- K = overall decay constant (T^{-1});
- k_b = decay constant for bulk flow reaction (T^{-1});
- k_f = mass-transfer coefficient (LT^{-1});
- k_w = decay constant for wall reaction (LT^{-1});
- L = pipe length (L);
- M = external mass inflow to junction (MT^{-1});
- O = set of pipes with flow out of junction;
- q = flow rate in pipe (L^3T^{-1});
- R = Reynolds number;
- r_h = hydraulic radius (L);
- S = external flow rate into junction (L^3T^{-1});
- Sc = Schmidt number;
- Sh = Sherwood number;
- t = time (T);
- u = flow velocity (LT^{-1});
- V = storage-tank volume (L^3);
- x = distance along pipe (L); and
- ν = kinematic viscosity (L^2T^{-1}).